Currency Converter

A

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Submitted to

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DECLARATION

We hereby declare that this Project Report titled “Simple Currency Converter Using Hardcoded Exchange Rate” submitted by us and approved by our project guide, to the Vivekananda Global University, Jaipur is a bonafide work undertaken by us and it is not submitted to any other University or Institution for the award of any degree diploma / certificate or published any time before.

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Simple Currency Converter Using Hardcoded Exchange Rates in Java

Abstract: A Simple Currency Converter in Java allows users to convert amounts between currencies using hardcoded exchange rates. It takes user input, performs the conversion, and displays the result. Since it doesn't rely on APIs, it works offline, but rates must be manually updated. Future improvements could include real-time rate integration.

1. Introduction: The Simple Currency Converter is a basic Java application designed. to convert one currency to another using predefined, hardcoded exchange rates. It offer quick and offline solution for currency conversion**.**

2. Objectives:

- Build a simple Java-based currency converter.
- Use hardcoded exchange rates for offline use.
- Practice basic Java concepts like input, output, and conditionals.
- Create a functional mini project for learning and demonstration.

3. System Requirements:

- **Hardware:** Any computer capable of running Java.
- **Software:**
  - Java Development Kit (JDK)
  - Integrated Development Environment (IDE) such as Eclipse or IntelliJ
  - Command Line or Terminal

4. System Design: The system consists of three main components:

1. **User Input Handling**
   - The system takes input from the user.
   - Source currency, target currency, and amount to be converted.
2. **Exchange Rate Mapping**
   - A predefined set of exchange rates is stored in the code using if-else.
3. **Conversion Process**
   - The program multiplies the input amount by the corresponding exchange rate.

4. **Implementation Details:**

```java
import java.util.Scanner;

public class CurrencyConverter {

    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);

        // Hardcoded exchange rates (1 INR to respective
            currencies)
        double usdRate = 0.012;   // 1 INR = 0.012 USD
        double eurRate = 0.011;   // 1 INR = 0.011 EUR
        double gbpRate = 0.0095;  // 1 INR = 0.0095 GBP
        double jpyRate = 1.62;    // 1 INR = 1.62 JPY

        System.out.println("----- Currency Converter (from INR)
            -----");
        System.out.print("Enter amount in INR: ");
        double inr = scanner.nextDouble();

        System.out.println("Choose currency to convert to:");
        System.out.println("1. USD");
```

```java
    System.out.println("2. EUR");
    System.out.println("3. GBP");
    System.out.println("4. JPY");
    System.out.print("Enter choice (1-4): ");
    int choice = scanner.nextInt();

    double result = 0;
    String currency = "";

    switch (choice) {
        case 1:
            result = inr * usdRate;
            currency = "USD";
            break;
        case 2:
            result = inr * eurRate;
            currency = "EUR";
            break;
        case 3:
            result = inr * gbpRate;
            currency = "GBP";
            break;
        case 4:
            result = inr * jpyRate;
            currency = "JPY";
            break;
        default:
            System.out.println("Invalid choice.");
            return;
    }

    System.out.printf("%.2f INR = %.2f %s\n", inr, result,
        currency);
    scanner.close();
}
```

- **Classes Used:**
  - `Currency Converter` (Main class Contain entire functionality)
  - `Currency Data` (Stores and provide access)
  - `Currency App` (Contains the main method)
- **Key Features:**
  - Hardcoded Exchange Rate
  - Multiple Currency Support
  - Simple Input and Output
  - Accurate Conversion Logic

6. Code Explanation: The conversion system is implemented using **Object-Oriented Programming (OOP)** principles. The main components of the system are:

6.1 Currency Connverter Class

This is the main class that handles the entire conversion process. It contains methods for:

- `start()`: Initiates the program loop with options for registration, login, and exiting.
- `convert(String fromCurrency, String toCurrency, double,Amount.`
- `loginUser()`: Authenticates users based on their credentials.
- `Get exchange rate()`: Return the map of supported currencies.
- isCurrencySupported(String currency).
- `castVote(int userId, int candidateId)`: Ensures a user can vote only once.
- initializeRates(): Initialize the hardcoded exchange rate.

6.2 Currency Data

This class defines the `data` object with attributes:

- getExchangeRate(String currency).
- isSupported(String currency).
- getAllCurrencies().
- getRateMap().

6.3 Currency App

This class defines the `Application` object with attributes:

- main(String[] args).
- displayMenu().
- getUserInput().
- displayResult(double result).

7. Sample Execution:

```
----- Currency Converter (from INR) -----
Enter amount in INR: 100
Choose currency to convert to:
1. USD
2. EUR
3. GBP
4. JPY
Enter choice (1-4): 1
100.00 INR = 1.20 USD
```

```
----- Currency Converter (from INR) -----
Enter amount in INR: 50
Choose currency to convert to:
1. USD
2. EUR
3. GBP
4. JPY
Enter choice (1-4): 2
50.00 INR = 0.55 EUR
```

7.1 Supported Currency:

```
USD, INR, EUR, GBP,
JPY
```

7.2 Source And Target:

```
Enter source currency: GBP

Enter target currency: JPY

Enter amount: 500
```

7.3 Results:

```
Conversion Results:

Supported Currencies: USD, INR, EUR, GBP, JPY

Enter source currency: GBP

Enter target currency: JPY

Enter amount: 500

500.00 GBP = 95696.20 JPY
```

8. Advantages and Disadvantages:

8.1 Advantages:

- Offline Functionality.
- Fast Execution.
- Easy to Implement.
- No Dependency on APIs.

8.2 Disadvantages:

- Outdated Rates.
- Limited Currency Support.
- No Real-Time Accuracy.
- Not Scalable.

9. Future Enhancements:

- Real-Time Exchange Rates via API.
- Graphical User Interface (GUI).
- Show recent conversion history and let users view past converted amounts.
- Display country flags and currency symbols (₹, $, €) for better presentation.
- Add support for multiple languages to make it usable across different regions.
- Convert the program into an Android app using Java and Android Studio.
- Add basic speech input and output using Java Speech APIs for accessibility.

10. Conclusion:

The Simple Currency Converter project demonstrates how basic Java concepts can be applied to solve real-world problems. By using hardcoded exchange rates, it allows users to perform quick and offline currency conversions. While the application is limited in scope, it serves as a strong foundation for understanding user input handling, conditionals, and data structures like HashMap. With further enhancements, this project can evolve into a more advanced, real-time currency converter.

11. References:

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